



ZYROO INTERNSHIP PROGRAM

DATA ANALYTICS INTERNSHIP

WEEK 3

Ride Analytics & Revenue Intelligence Platform

Task 03: Revenue & Driver Performance Analysis

Difficulty	Beginner to Intermediate
Category	Data Analytics, Revenue Analytics, Driver Analytics
Estimated Time	6 to 10 hours

Project Progress

Week 2: We analyzed ride demand by date, month, hour, weekday/weekend, and location.

Customer Analysis: We identified customer frequency, new vs repeat customers, and important customer behavior patterns.

Current Stage: We now use these demand and customer insights to analyze revenue and driver performance.

Goal

Understand where revenue comes from and how drivers perform. Use the same ride-hailing dataset to analyze the financial and operational side of the business.

1. Understand the Business Problem

A ride-hailing company needs to know more than the number of rides it receives. Management needs to understand revenue sources, driver productivity, ratings, completion, and cancellations.

- How much revenue is generated?
- Which locations generate the most revenue?
- Which payment methods are most common?
- Which ride types generate more revenue?
- Which drivers complete the most rides?
- Which drivers generate the most revenue?

- Which drivers have strong ratings?
- Which drivers have high cancellation rates?

2. Dataset Requirements

Continue using the same ride-hailing dataset used in Week 2.

```
ride_id date
pickup_location
dropoff_location
distance fare
payment_method
driver_id
rating
ride_status
```

If available, you can also use `ride_type`, `vehicle_type`, `driver_rating`, `pickup_time`, and `dropoff_time`.

3. Data Preparation

Before starting the revenue and driver analysis, check the required fields.

- Fare values
- Ride status
- Driver IDs
- Ratings
- Distance
- Payment method
- Ride type
- Missing values
- Duplicate records
- Unusual fare values

For revenue analysis, clearly define which rides count as completed revenue.

```
Completed Ride → Revenue counted
Cancelled Ride → Revenue normally not counted
```

Follow the rules of your selected dataset if it defines revenue differently.

4. Revenue Analysis

Calculate and analyze the total revenue generated by the ride-hailing business.

Analyze:

- Total revenue
- Revenue by month
- Revenue by day
- Revenue by location
- Revenue by payment method
- Revenue by ride type
- Average fare
- Average revenue per completed ride

5. Total Revenue

Calculate the total revenue from completed rides.

```
Total Completed Rides = 8,500
Total Revenue = PKR 6,750,000
```

Also calculate average revenue per completed ride.

```
Average Revenue per Ride =
Total Revenue / Completed Rides
```

6. Monthly Revenue Analysis

Analyze how revenue changes over time. Create a line or column chart and identify the highest and lowest revenue periods.

7. Revenue by Location

Analyze revenue generated by pickup locations.

- Highest-revenue location
- Lowest-revenue location
- Top 10 revenue-generating locations
- Revenue contribution of major locations

8. Revenue by Payment Method

Analyze revenue based on payment method.

- Revenue by payment method
- Number of completed rides by payment method
- Average fare by payment method

Questions:

- Which payment method generates the most revenue?
- Which payment method is used most frequently?
- Does the highest number of rides also produce the highest revenue?

9. Revenue by Ride Type

If your dataset contains ride types or vehicle types, compare their usage and revenue.

- Most-used ride type
- Highest-revenue ride type
- Average fare by ride type

Example:

Ride Type	Rides	Revenue
Economy	4,500	PKR 2.8M
Standard	2,800	PKR 2.2M
Premium	1,200	PKR 1.7M

10. Driver Performance Analysis

Analyze individual driver performance to identify high-performing drivers, low-performing drivers, high-revenue drivers, highly rated drivers, and drivers with high cancellation rates.

11. Driver KPIs

Rides per Driver: Number of rides handled by each driver.

Completed Rides: Number of successfully completed rides.

Revenue: Total revenue generated from completed rides.

Average Rating: Average customer rating received by the driver.

Completion Rate:

```
Completion Rate =  
Completed Rides / Total Rides × 100
```

Cancellation Rate:

```
Cancellation Rate =  
Cancelled Rides / Total Rides × 100
```

Average Distance: Average distance covered by each driver.

12. Driver Performance Table

Driver	Rides	Revenue	Rating	Completion	Cancellation
Driver A	245	PKR 185K	4.8	96%	4%
Driver B	221	PKR 164K	4.6	92%	8%
Driver C	198	PKR 151K	4.7	94%	6%

These are example values only. Use actual values from your dataset.

13. Driver Performance Ranking

Create a ranking of drivers. Don't rank drivers using revenue alone. Consider completed rides, revenue, rating, completion rate, and cancellation rate.

Rank	Driver	Revenue	Rating	Completion
1	Driver A	PKR 185K	4.8	96%
2	Driver C	PKR 151K	4.7	94%
3	Driver B	PKR 164K	4.6	92%

Your ranking method should be explained clearly.

14. SQL Analysis Practice

Use SQL to analyze revenue and driver performance. Adapt the syntax to your database.

Revenue by Payment Method

```
SELECT payment_method,  
COUNT(*) AS completed_rides,  
SUM(fare) AS total_revenue,  
AVG(fare) AS average_fare  
FROM rides  
WHERE ride_status = 'Completed'  
GROUP BY payment_method  
ORDER BY total_revenue DESC;
```

Driver Performance

```
SELECT driver_id,  
COUNT(*) AS total_rides,  
SUM(CASE WHEN ride_status = 'Completed' THEN 1 ELSE 0 END) AS completed_rides,  
SUM(CASE WHEN ride_status = 'Cancelled' THEN 1 ELSE 0 END) AS cancelled_rides,  
SUM(CASE WHEN ride_status = 'Completed' THEN fare ELSE 0 END) AS revenue,  
AVG(rating) AS average_rating  
FROM rides  
GROUP BY driver_id  
ORDER BY revenue DESC;
```

15. Python / Pandas Analysis Practice Total Revenue

```
completed = df[df["ride_status"] == "Completed"]  
  
total_revenue = completed["fare"].sum()  
average_fare = completed["fare"].mean()
```

Revenue by Location

```
revenue_by_location = (  
    completed.groupby("pickup_location")["fare"]  
    .sum()  
    .sort_values(ascending=False)  
)
```

Driver Performance

```
driver_summary = (  
    df.groupby("driver_id")  
    .agg(  
        rides=("ride_id", "count"),  
        revenue=("fare", "sum"),  
        rating=("rating", "mean"),  
        average_distance=("distance", "mean") )  
    .sort_values("revenue", ascending=False)  
)
```

16. Visualization Tasks

Create clear visualizations for the revenue and driver analysis.

Revenue Charts

- Revenue by month
- Revenue by location
- Revenue by payment method
- Revenue by ride type
- Average fare by ride type

Driver Charts

- Top drivers by revenue
- Top drivers by completed rides
- Driver rating comparison
- Driver completion rate
- Driver cancellation rate

17. Business Questions

- What is the total revenue?
- Which month generated the highest revenue?
- Which location generates the most revenue?
- Which payment method contributes the most revenue?
- Which ride type generates the most revenue?
- What is the average fare?
- Which drivers generate the most revenue?
- Which drivers complete the most rides? ■ Which drivers have the highest ratings?
- Which drivers have high cancellation rates?
- Do high-revenue drivers also have strong completion rates?
- Are there locations with high demand but relatively low revenue?

18. Business Insights

Convert your analysis into meaningful business findings. Use actual results from the dataset.

Example: Driver A generated the highest revenue and also maintained a high completion rate, making the driver a strong contributor to business performance.

Example: A location has high ride demand but relatively low revenue, which may indicate shorter trips or a lower-value ride mix.

Example: A driver has a high cancellation rate, which should be investigated because frequent cancellations can affect customer experience and completed-ride revenue.

19. Business Recommendations

Based on your findings, provide practical recommendations.

- Provide incentives to high-performing drivers.
- Investigate drivers with unusually high cancellation rates.
- Increase driver availability in high-revenue locations.
- Monitor locations with high demand but low revenue.
- Promote useful digital payment options where appropriate.
- Review ride types that generate strong revenue.
- Use driver performance data to improve driver allocation.

20. Week 3 Deliverable

Prepare a **Revenue & Driver Performance Analysis Report**.

Your report should contain:

- Business problem
- Dataset
- Data preparation
- Total and monthly revenue
- Revenue by location
- Revenue by payment method
- Revenue by ride type, if available
- Average fare
- Driver performance metrics
- Driver ranking
- Visualizations
- At least 5 key findings
- Business insights
- At least 3 recommendations
- Limitations

21. GitHub Submission

Continue using the same project repository.

```
ride-analytics-revenue-intelligence/
```

■

```
■■■ data/
■■■ sql/
■■■ python/
■ ■■■ week-02-demand-customer-analysis/
■ ■■■ week-03-revenue-driver-analysis/
■■■ powerbi/
■■■ reports/
■ ■■■ week-02/
■ ■■■ week-03/
■■■ screenshots/
■■■ README.md
■■■ requirements.txt
```

Your Week 3 folder should contain:

- Python notebook/scripts
- SQL queries
- Revenue analysis
- Driver analysis
- Charts
- Report
- Screenshots

22. README Update

- Week 3 objective
- Revenue KPIs
- Driver KPIs
- Revenue findings
- Driver findings
- Charts
- Business insights
- Recommendations
- Limitations

Keep Week 2 and Week 3 results connected because they are part of the same project.

23. Submission Checklist

- Total revenue calculated
- Average fare calculated
- Monthly revenue analyzed
- Revenue by location analyzed

- Revenue by payment method analyzed
- Revenue by ride type analyzed, if available
- Driver performance analyzed
- Completed rides calculated
- Driver revenue calculated
- Average driver rating calculated
- Completion rate calculated
- Cancellation rate calculated
- Driver ranking created
- Revenue charts created
- Driver charts created
- At least 5 key findings written
- Business insights written
- Recommendations prepared
- GitHub repository updated
- README updated

■ Week 3 Success Criteria

By the end of Week 3, you should be able to explain:

How much revenue does the business generate?
Where does most of the revenue come from?
Which drivers perform best?
Which drivers need attention?
What actions should the business take based on the data?

Final Flow

Demand & Customer Analysis → Revenue Analysis → Driver Performance → Performance Ranking → Business Insights

■ Next Stage: Week 4

Week 4 will move from analysis to Business Intelligence and Dashboard Development.

You will take the Week 2 demand insights + Week 3 revenue and driver insights and turn them into an interactive Power BI/Tableau Ride Analytics Dashboard with KPI cards, filters, charts, and business-focused views.

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